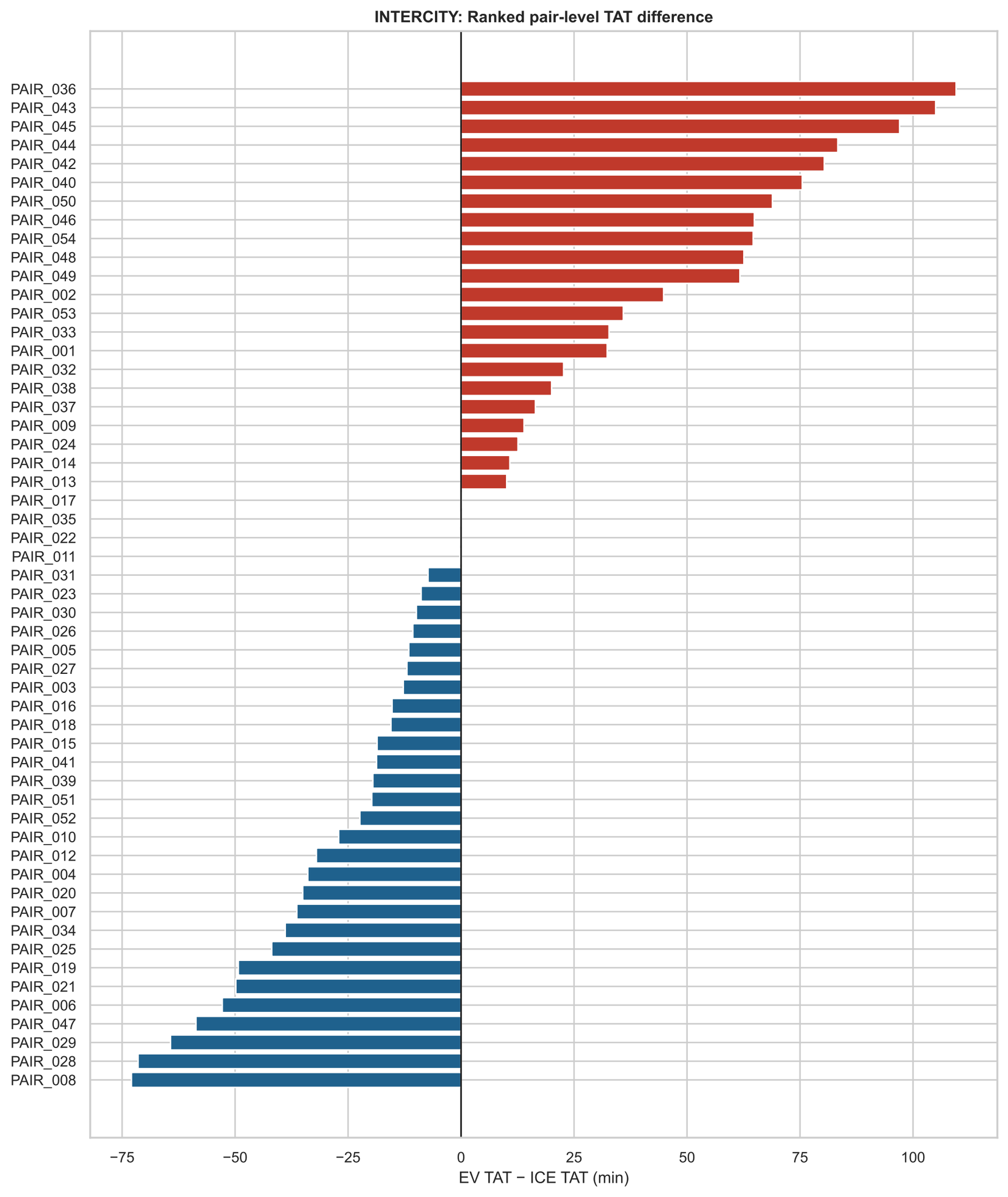
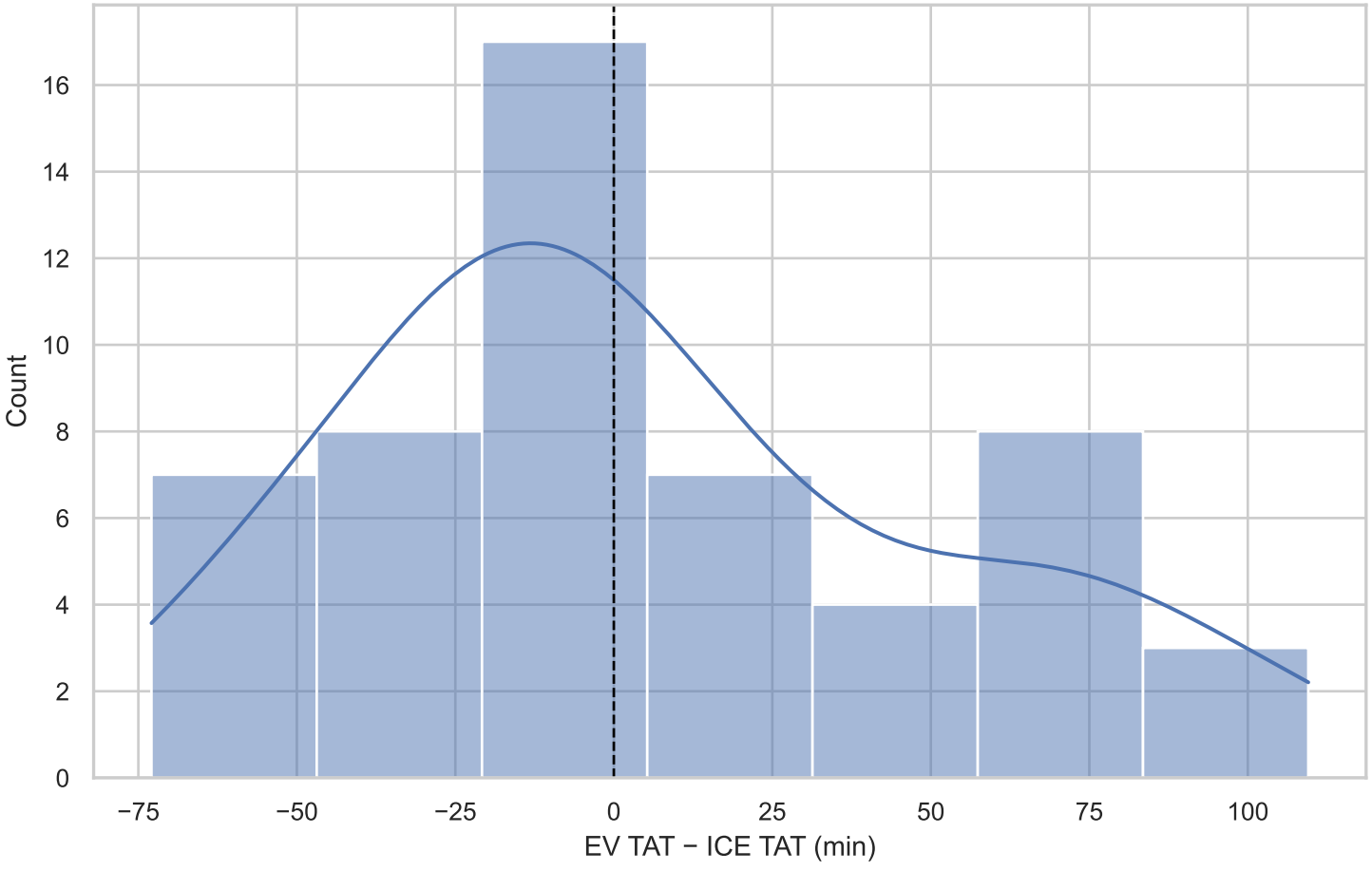
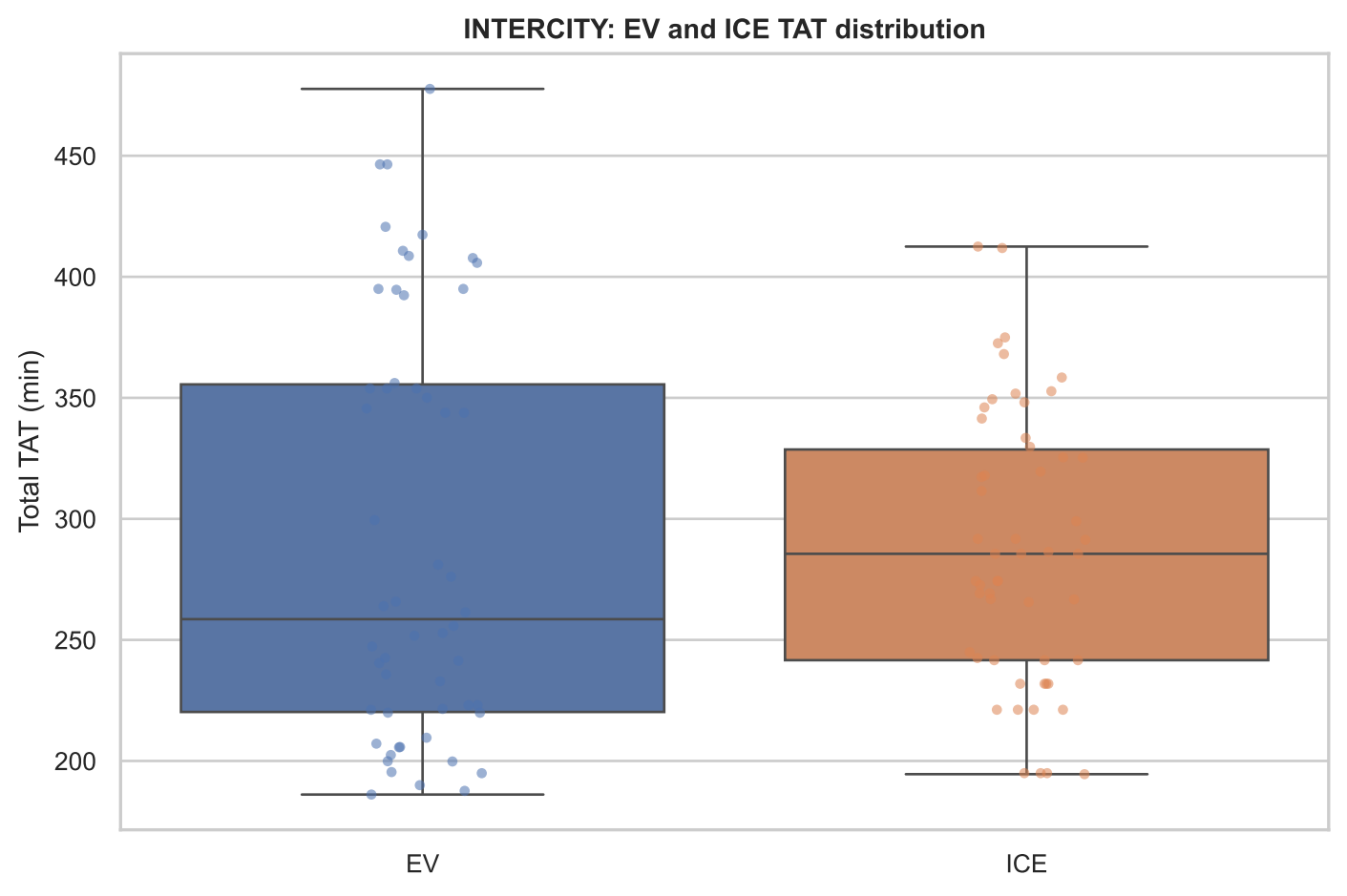
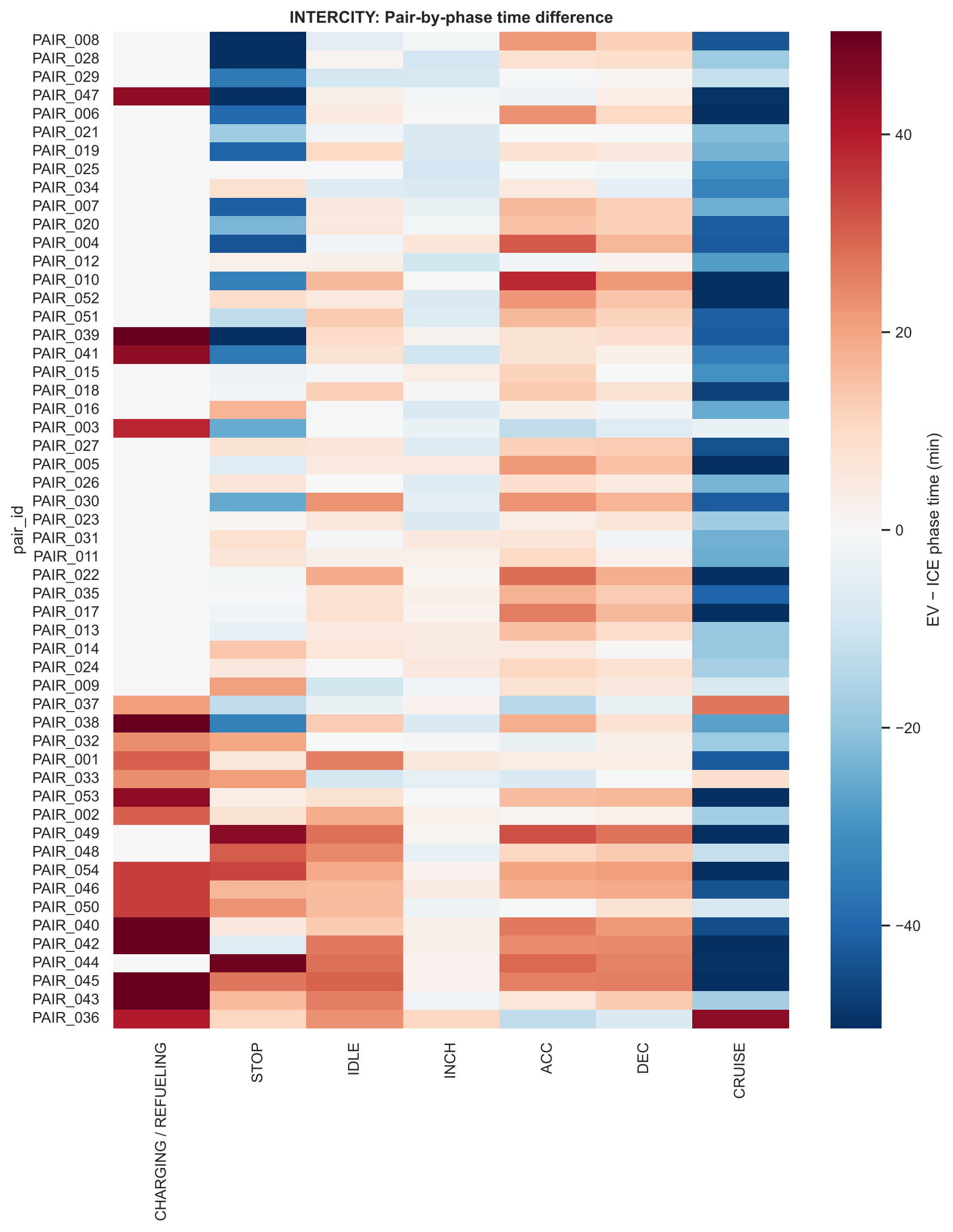
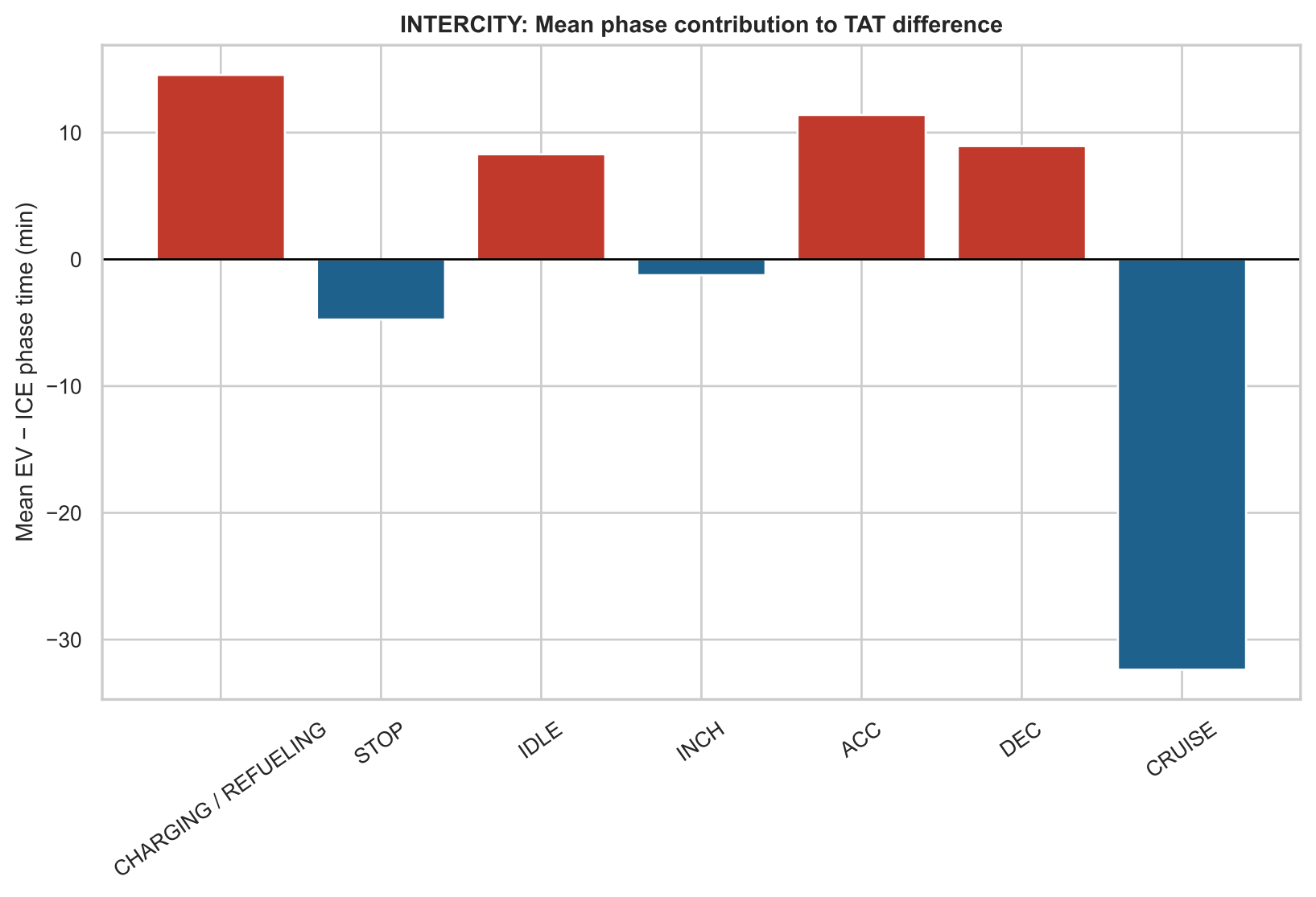
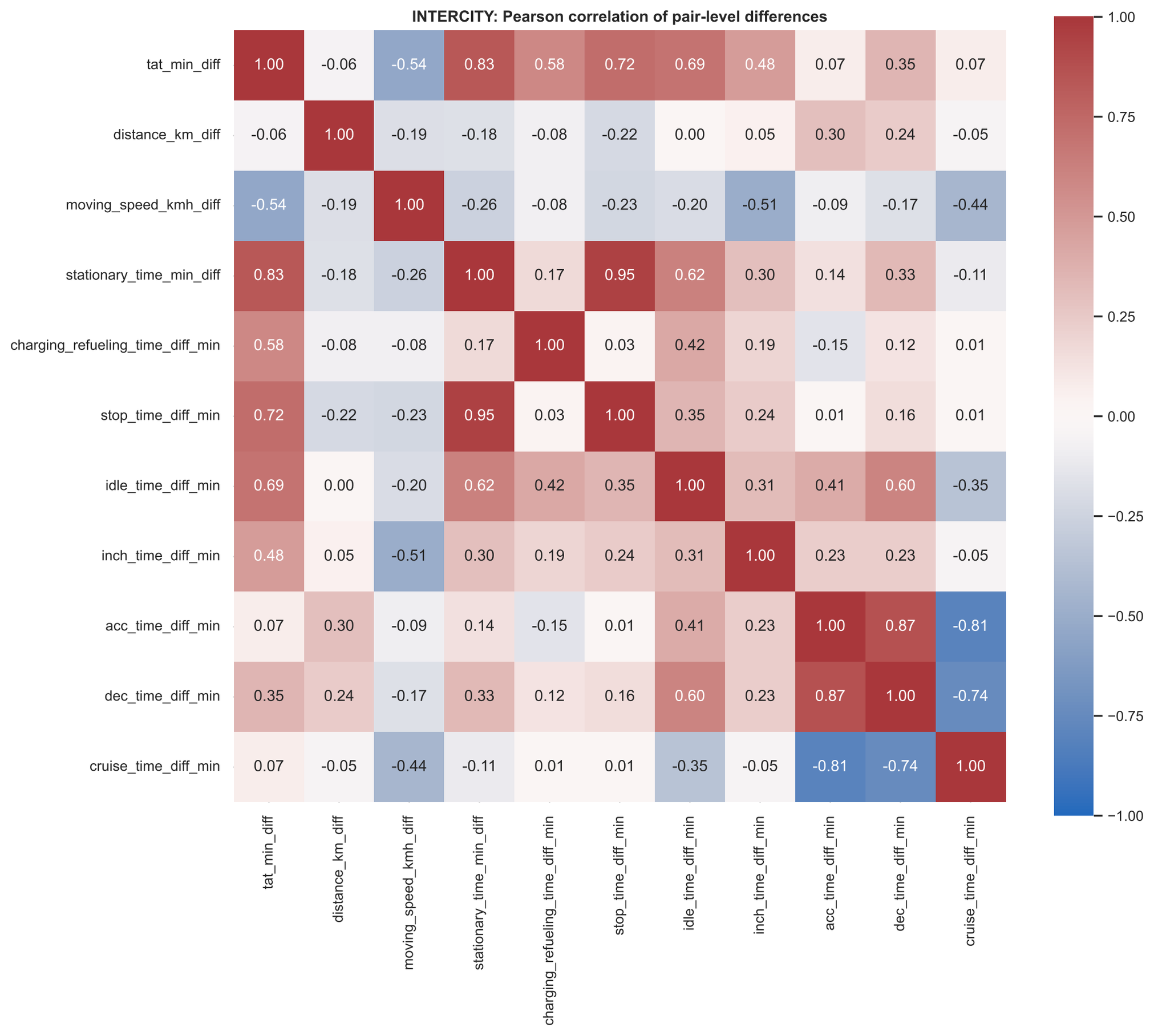

INTERCITY: Ranked pair-level TAT difference
PAIR_036
PAIR_043
PAIR_045
PAIR_044
PAIR_042
PAIR_040
PAIR_050
PAIR_046
PAIR_054
PAIR_048
PAIR_049
PAIR_002
PAIR_053
PAIR_033
PAIR_001
PAIR_032
PAIR_038
PAIR_037
PAIR_009
PAIR_024
PAIR_014
PAIR_013
PAIR_017
PAIR_035
PAIR_022
PAIR_011
PAIR_031
PAIR_023
PAIR_030
PAIR_026
PAIR_005
PAIR_027
PAIR_003
PAIR_016
PAIR_018
PAIR_015
PAIR_041
PAIR_039
PAIR_051
PAIR_052
PAIR_010
PAIR_012
PAIR_004
PAIR_020
PAIR_007
PAIR_034
PAIR_025
PAIR_019
PAIR_021
PAIR_006
PAIR_047
PAIR_029
PAIR_028
PAIR_008
−75
−50
−25
0
25
50
75
100
EV TAT − ICE TAT (min)

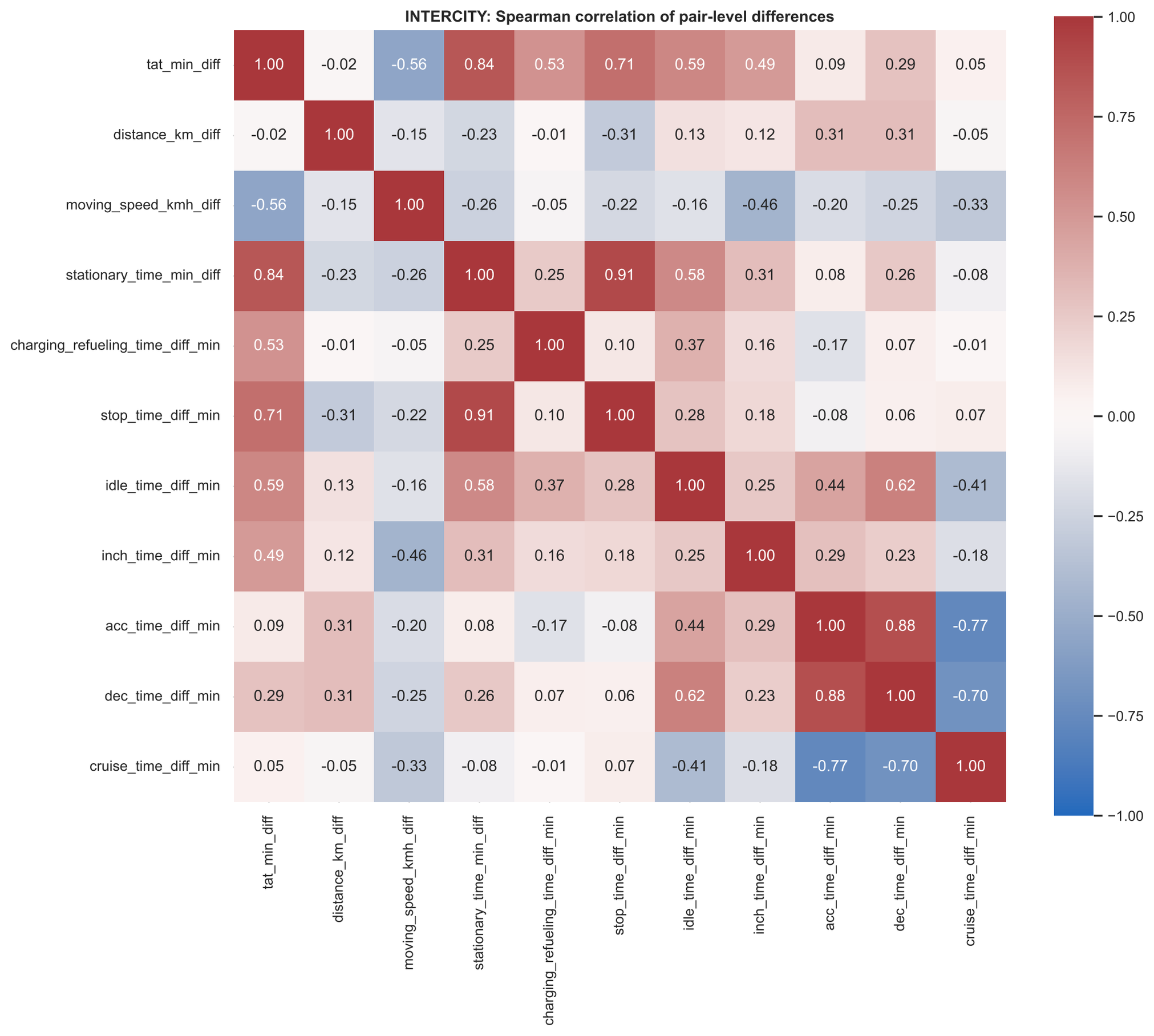

INTERCITY: Distribution of TAT differences
Count
0
2
4
6
8
10
12
14
16
−75
−50
−25
0
25
50
75
100
EV TAT − ICE TAT (min)

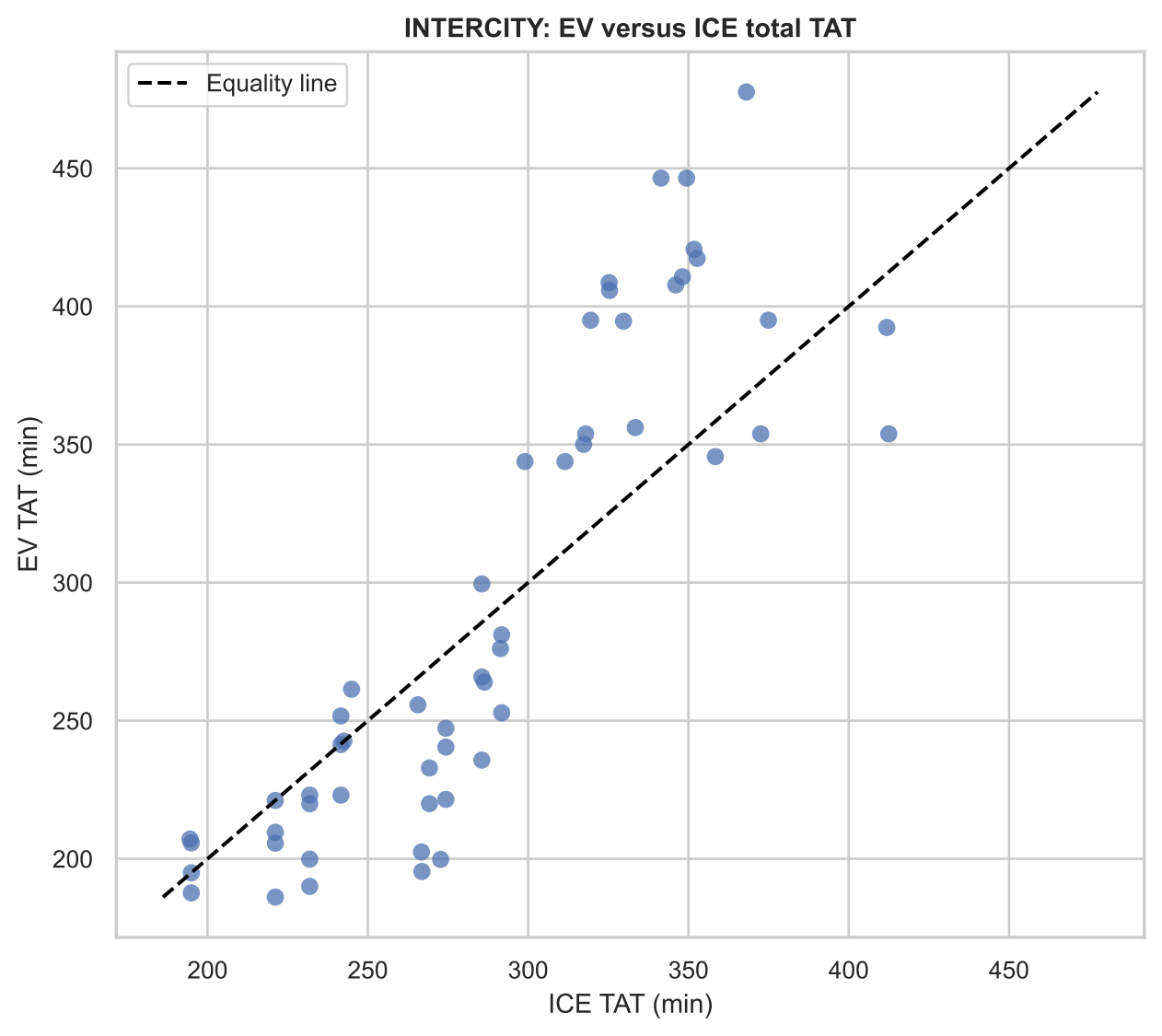

INTERCITY: EV and ICE TAT distribution
Total TAT (min)
450
400
350
300
250
200
EV
ICE

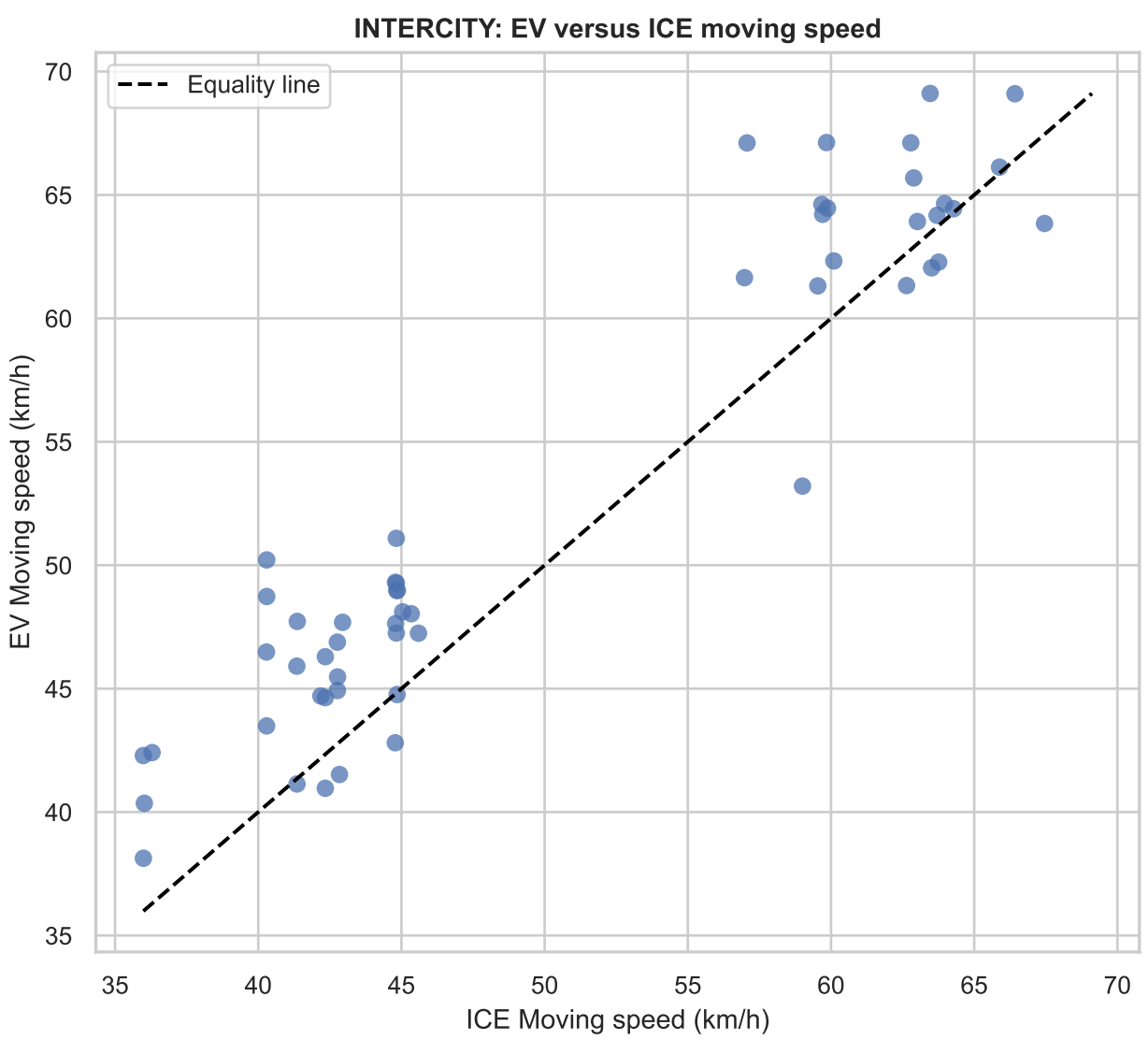

INTERCITY: Pair-by-phase time difference
pair_id
PAIR_008
PAIR_028
PAIR_029
PAIR_047
PAIR_006
PAIR_021
PAIR_019
PAIR_025
PAIR_034
PAIR_007
PAIR_020
PAIR_004
PAIR_012
PAIR_010
PAIR_052
PAIR_051
PAIR_039
PAIR_041
PAIR_015
PAIR_018
PAIR_016
PAIR_003
PAIR_027
PAIR_005
PAIR_026
PAIR_030
PAIR_023
PAIR_031
PAIR_011
PAIR_022
PAIR_035
PAIR_017
PAIR_013
PAIR_014
PAIR_024
PAIR_009
PAIR_037
PAIR_038
PAIR_032
PAIR_001
PAIR_033
PAIR_053
PAIR_002
PAIR_049
PAIR_048
PAIR_054
PAIR_046
PAIR_050
PAIR_040
PAIR_042
PAIR_044
PAIR_045
PAIR_043
PAIR_036
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
20
0
−20
−40

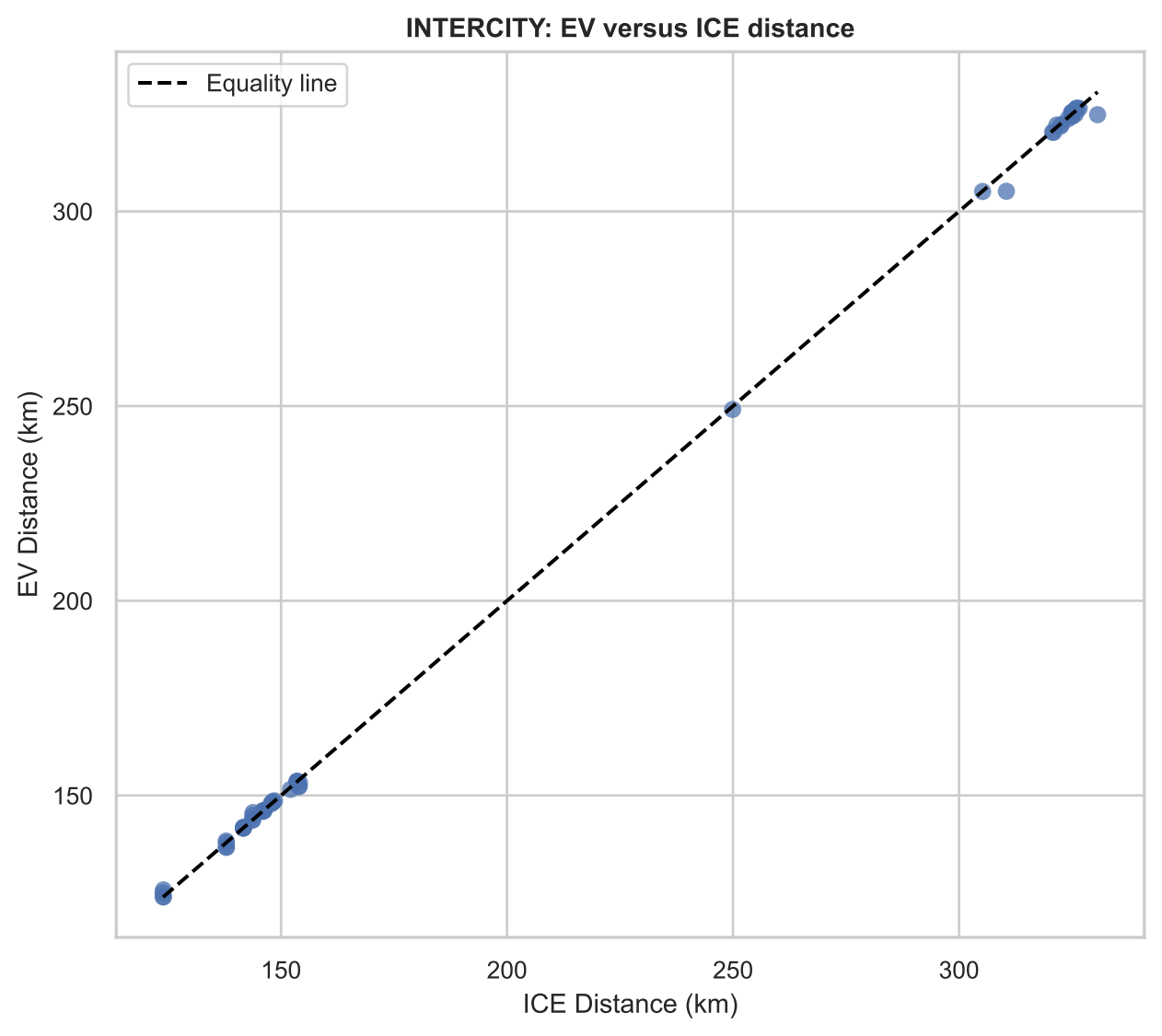

INTERCITY: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
10
0
−10
−20
−30
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

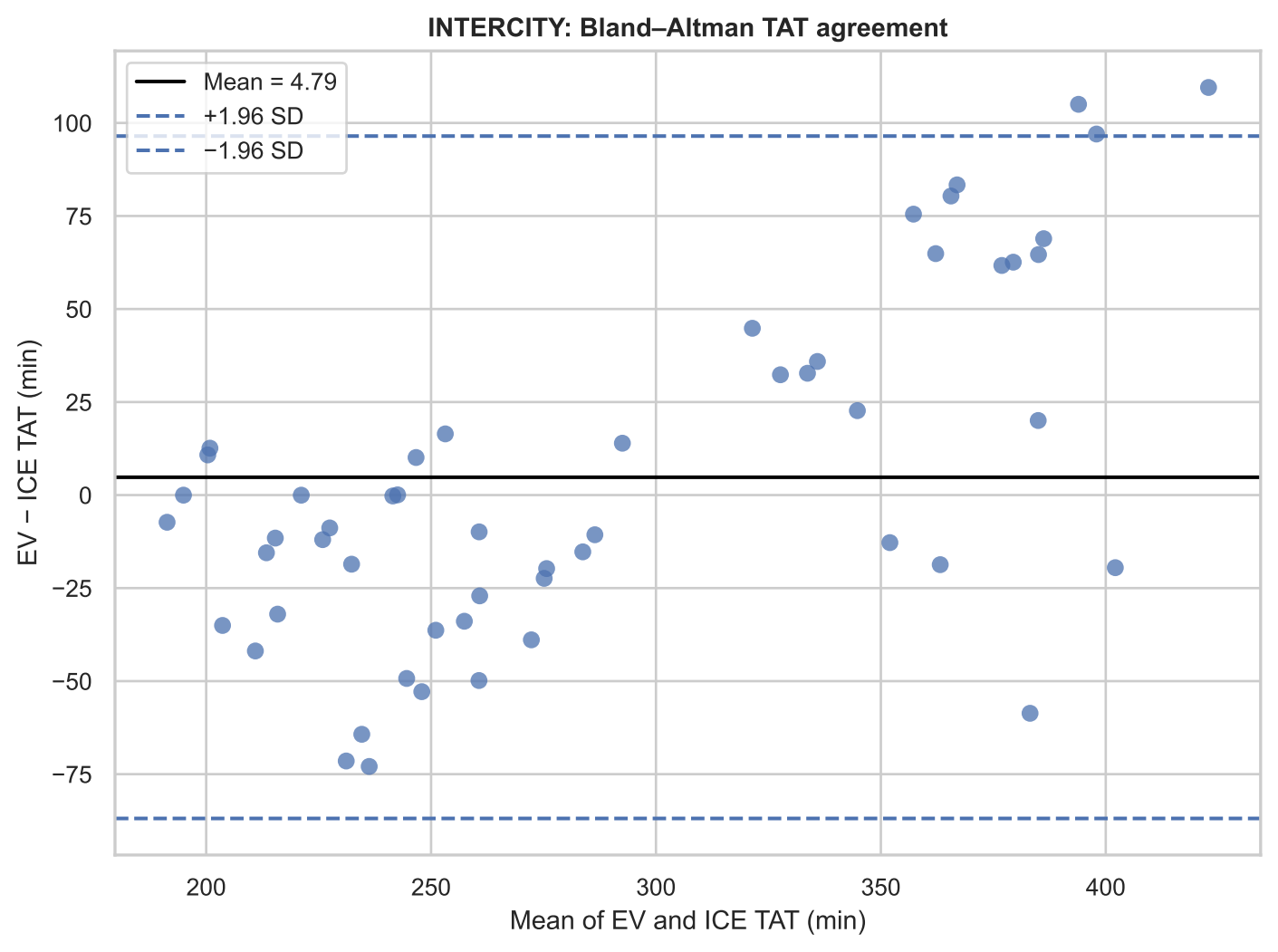

INTERCITY: Pearson correlation of pair-level differences
tat_min_diff 1.00 -0.06 -0.54 0.83 0.58 0.72 0.69 0.48 0.07 0.35 0.07
distance_km_diff -0.06 1.00 -0.19 -0.18 -0.08 -0.22 0.00 0.05 0.30 0.24 -0.05
moving_speed_kmh_diff -0.54 -0.19 1.00 -0.26 -0.08 -0.23 -0.20 -0.51 -0.09 -0.17 -0.44
stationary_time_min_diff 0.83 -0.18 -0.26 1.00 0.17 0.95 0.62 0.30 0.14 0.33 -0.11
charging_refueling_time_diff_min 0.58 -0.08 -0.08 0.17 1.00 0.03 0.42 0.19 -0.15 0.12 0.01
stop_time_diff_min 0.72 -0.22 -0.23 0.95 0.03 1.00 0.35 0.24 0.01 0.16 0.01
idle_time_diff_min 0.69 0.00 -0.20 0.62 0.42 0.35 1.00 0.31 0.41 0.60 -0.35
inch_time_diff_min 0.48 0.05 -0.51 0.30 0.19 0.24 0.31 1.00 0.23 0.23 -0.05
acc_time_diff_min 0.07 0.30 -0.09 0.14 -0.15 0.01 0.41 0.23 1.00 0.87 -0.81
dec_time_diff_min 0.35 0.24 -0.17 0.33 0.12 0.16 0.60 0.23 0.87 1.00 -0.74
cruise_time_diff_min 0.07 -0.05 -0.44 -0.11 0.01 0.01 -0.35 -0.05 -0.81 -0.74 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

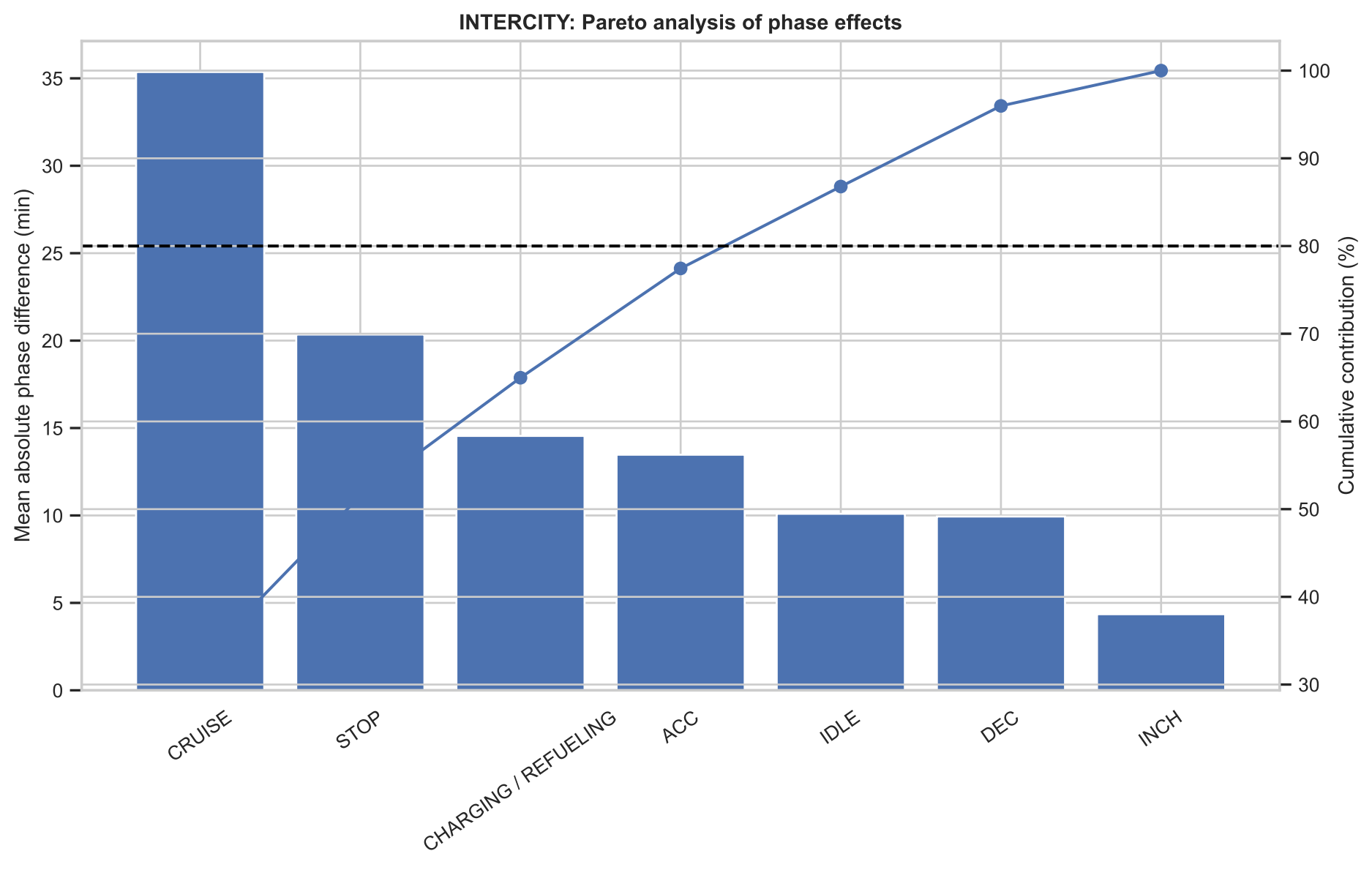

INTERCITY: Spearman correlation of pair-level differences
tat_min_diff 1.00 -0.02 -0.56 0.84 0.53 0.71 0.59 0.49 0.09 0.29 0.05
distance_km_diff -0.02 1.00 -0.15 -0.23 -0.01 -0.31 0.13 0.12 0.31 0.31 -0.05
moving_speed_kmh_diff -0.56 -0.15 1.00 -0.26 -0.05 -0.22 -0.16 -0.46 -0.20 -0.25 -0.33
stationary_time_min_diff 0.84 -0.23 -0.26 1.00 0.25 0.91 0.58 0.31 0.08 0.26 -0.08
charging_refueling_time_diff_min 0.53 -0.01 -0.05 0.25 1.00 0.10 0.37 0.16 -0.17 0.07 -0.01
stop_time_diff_min 0.71 -0.31 -0.22 0.91 0.10 1.00 0.28 0.18 -0.08 0.06 0.07
idle_time_diff_min 0.59 0.13 -0.16 0.58 0.37 0.28 1.00 0.25 0.44 0.62 -0.41
inch_time_diff_min 0.49 0.12 -0.46 0.31 0.16 0.18 0.25 1.00 0.29 0.23 -0.18
acc_time_diff_min 0.09 0.31 -0.20 0.08 -0.17 -0.08 0.44 0.29 1.00 0.88 -0.77
dec_time_diff_min 0.29 0.31 -0.25 0.26 0.07 0.06 0.62 0.23 0.88 1.00 -0.70
cruise_time_diff_min 0.05 -0.05 -0.33 -0.08 -0.01 0.07 -0.41 -0.18 -0.77 -0.70 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

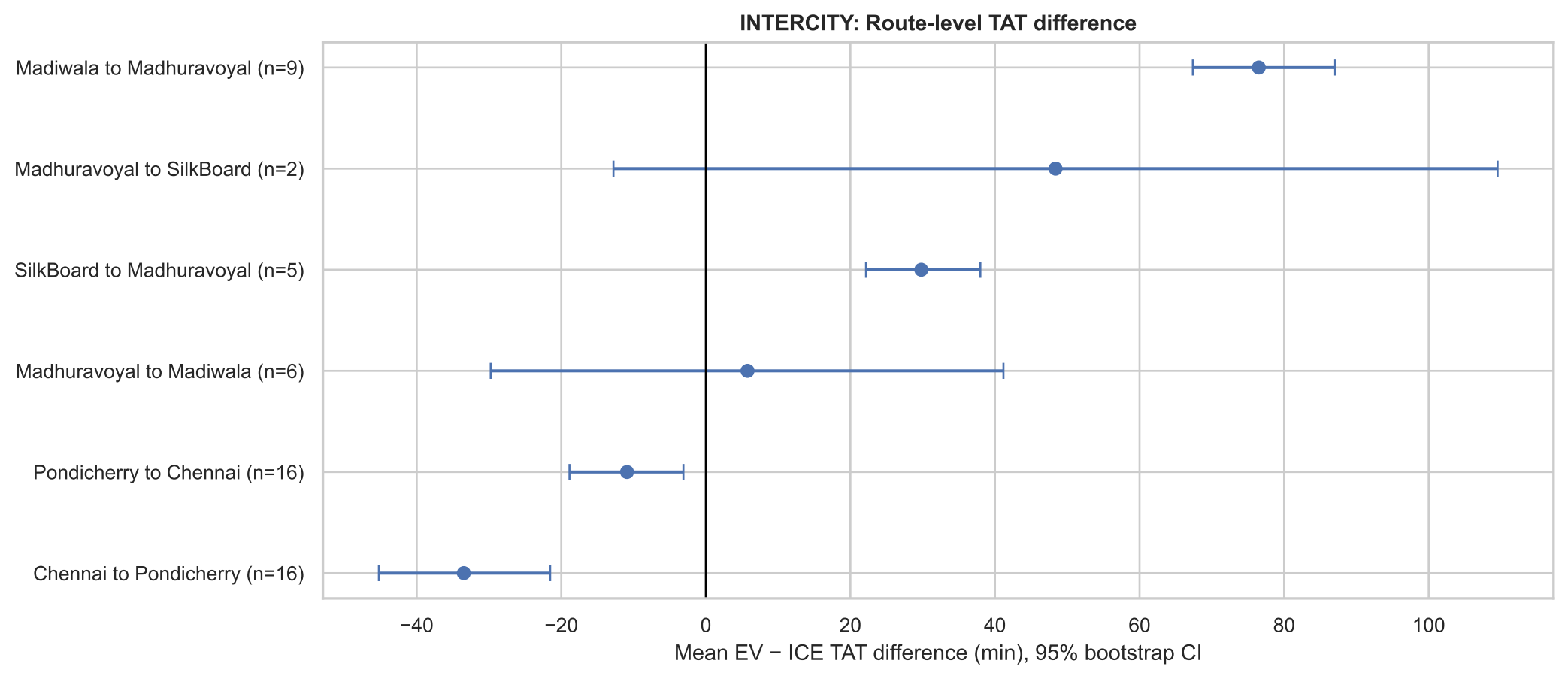

INTERCITY: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
200
250
300
350
400
450

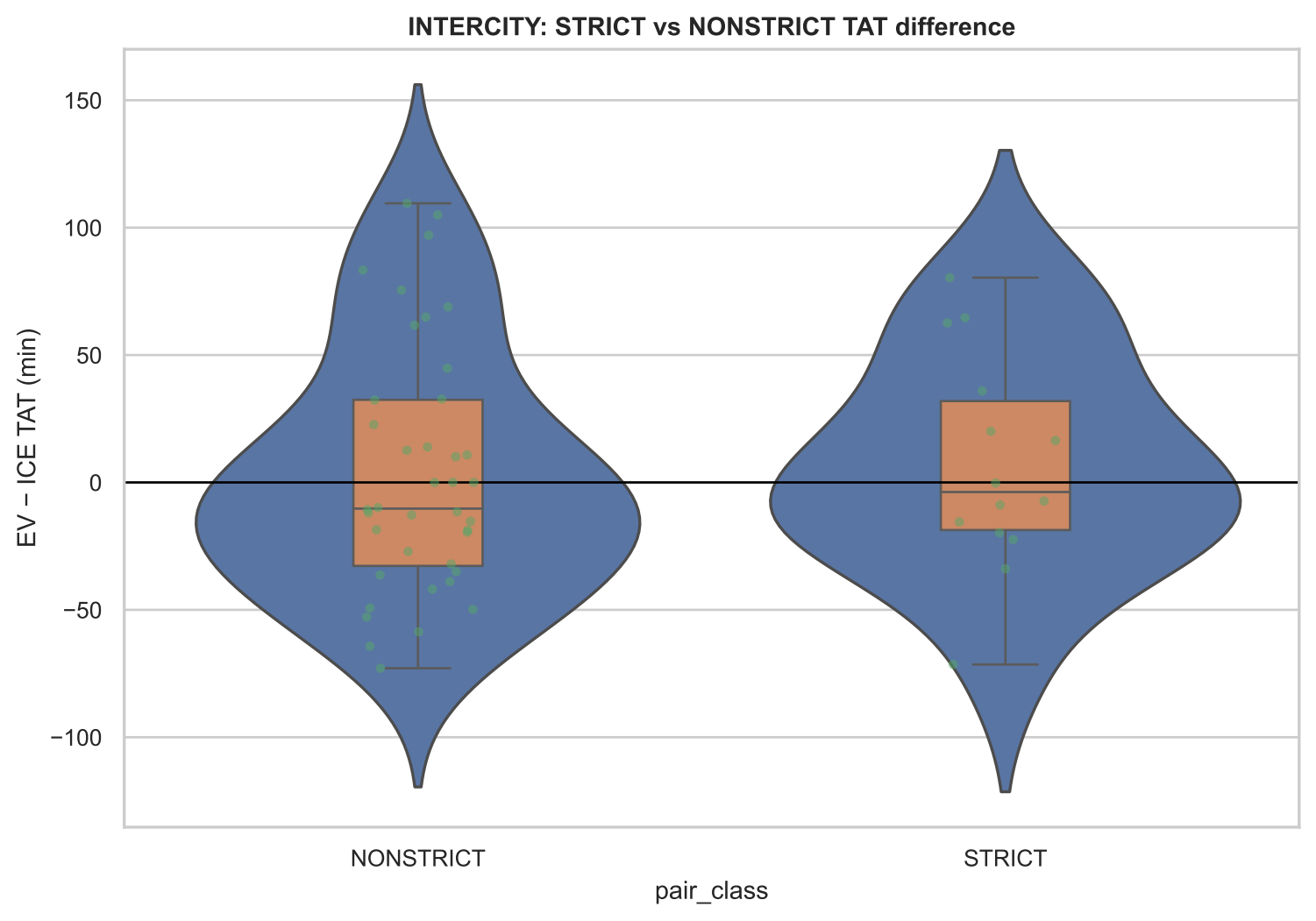

INTERCITY: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
70
65
60
55
50
45
40
35
35
40
45
50
55
60
65
70

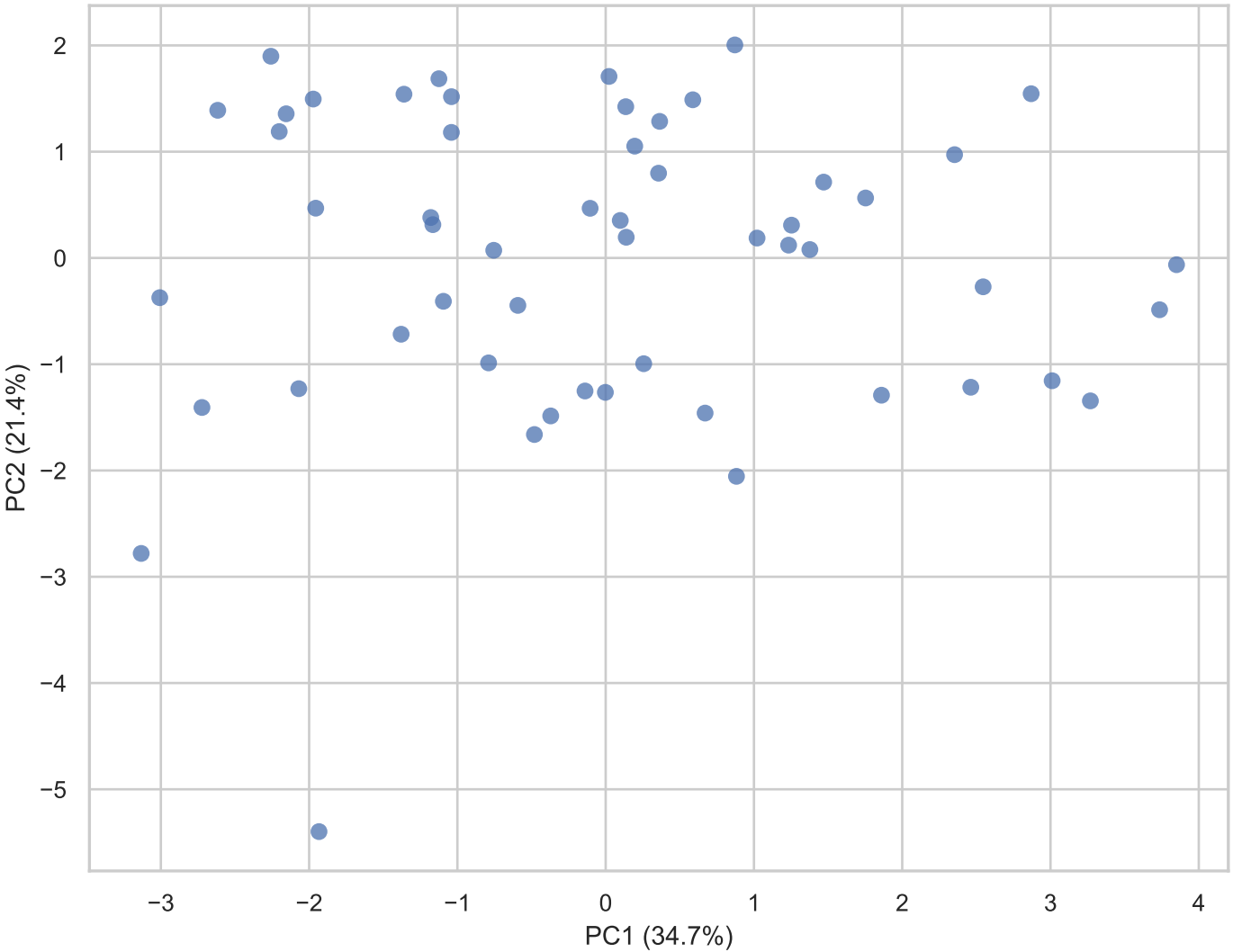

INTERCITY: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
150
200
250
300
150
200
250
300

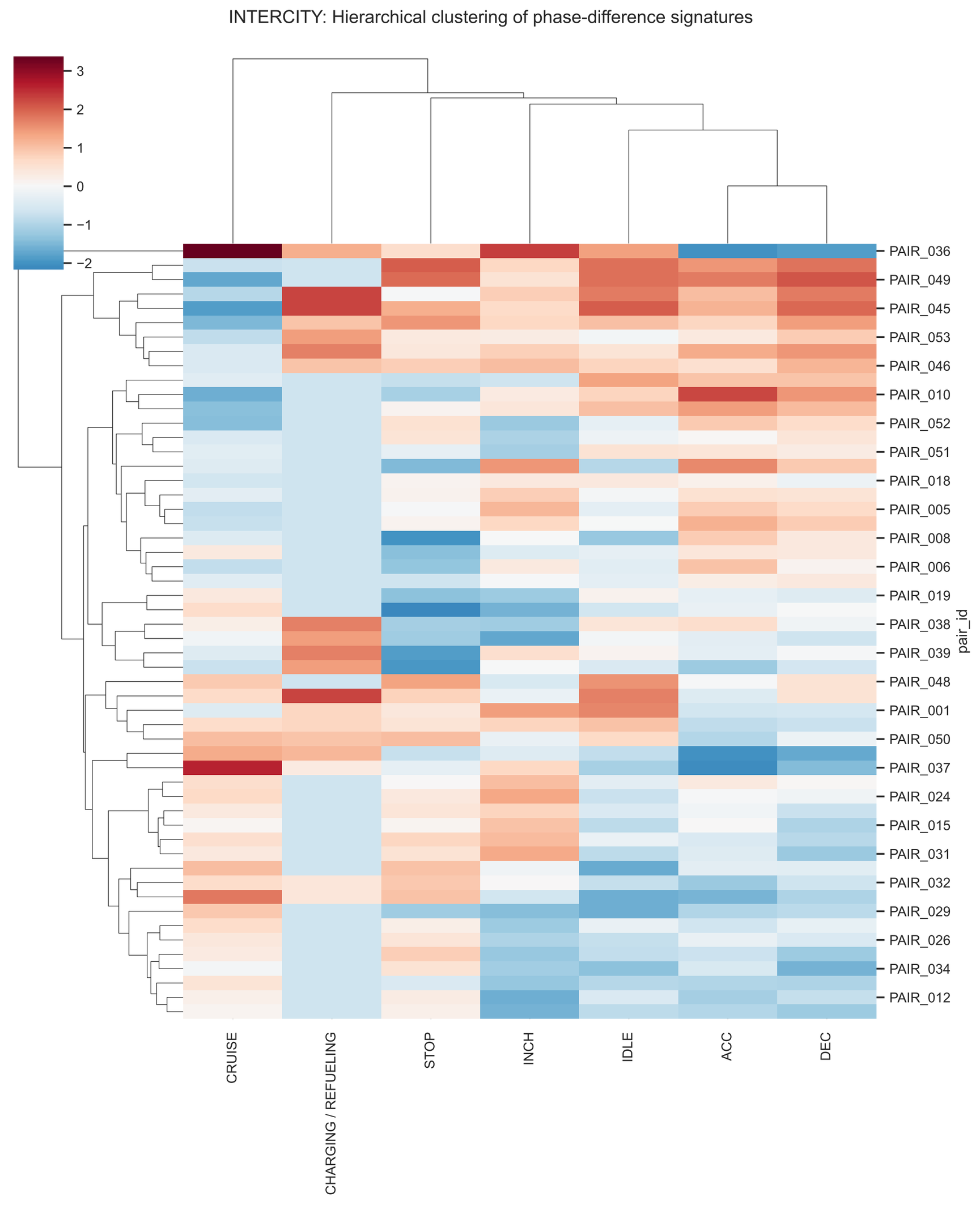

INTERCITY: Bland–Altman TAT agreement
Mean = 4.79
+1.96 SD
−1.96 SD
EV − ICE TAT (min)
Mean of EV and ICE TAT (min)
100
75
50
25
0
−25
−50
−75
200
250
300
350
400


INTERCITY: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
35
30
40
50
60
70
80
90
100
CRUISE
STOP
CHARGING / REFUELING
ACC
IDLE
DEC
INCH


INTERCITY: Route-level TAT difference
Madiwala to Madhuravoyal (n=9)
Madhuravoyal to SilkBoard (n=2)
SilkBoard to Madhuravoyal (n=5)
Madhuravoyal to Madiwala (n=6)
Pondicherry to Chennai (n=16)
Chennai to Pondicherry (n=16)
−40
−20
0
20
40
60
80
100
Mean EV − ICE TAT difference (min), 95% bootstrap CI


INTERCITY: STRICT vs NONSTRICT TAT difference
150
100
50
0
−50
−100
EV – ICE TAT (min)
NONSTRICT
STRICT
pair_class


INTERCITY: PCA of EV−ICE operational differences
PC2 (21.4%)
PC1 (34.7%)
2
1
0
−1
−2
−3
−4
−5
−3
−2
−1
0
1
2
3
4


INTERCITY: Hierarchical clustering of phase-difference signatures
3
2
1
0
−1
−2
PAIR_036
PAIR_049
PAIR_045
PAIR_053
PAIR_046
PAIR_010
PAIR_052
PAIR_051
PAIR_018
PAIR_005
PAIR_008
PAIR_006
PAIR_019
PAIR_038
PAIR_039
PAIR_048
PAIR_001
PAIR_050
PAIR_037
PAIR_024
PAIR_015
PAIR_031
PAIR_032
PAIR_029
PAIR_026
PAIR_034
PAIR_012
pair_id
CRUISE
CHARGING / REFUELING
STOP
INCH
IDLE
ACC
DEC